Enhancing Recruitment Decisions through Culture Fit Assessment: A Hybrid Approach Using Cosine Similarity (BERT Model) and Generative AI

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***Abstract*—In the contemporary recruitment landscape, assessing a candidate's alignment with organizational culture, known as "culture fit," has become as critical as evaluating technical qualifications. This paper presents a hybrid approach that combines machine learning and generative AI to analyze culture fit between candidates and job descriptions, enabling data-driven insights into hiring decisions. By leveraging cosine similarity with a BERT-based model and integrating Google's Generative AI for nuanced assessment, we establish a dual-model framework to evaluate cultural alignment. This methodology not only automates and scales culture fit assessment but also enhances accuracy by examining job descriptions, resumes, and behavioral responses holistically. Data for this study was sourced from Kaggle, facilitating the analysis of diverse job descriptions and resumes. We provide an in-depth evaluation of each model's efficacy, illustrating how this hybrid approach refines recruitment practices by offering tailored insights into candidate compatibility with organizational culture. The findings underscore the potential of AI-driven methods in transforming traditional HR processes, contributing to more informed, effective, and inclusive hiring decisions.**

***Index Terms*—Culture Fit Assessment, Artificial Intelligence (AI), Machine Learning, Natural Language Processing (NLP), Generative AI, BERT Model, Cosine Similarity, Recruitment Automation, Human Resources (HR), Employee Retention, Behavioral Analysis, Candidate Screening, Generative Language Models, Job Description Analysis, Data-Driven Hiring**

I. Introduction

For any company or organization, selecting the best applicant for a position is one of the most essential responsibilities. Though experience and technical proficiency are important, they are not the only criteria that determine a candidate's likelihood of success in a position. The extent to which a candidate fits in with the company's values, culture, and workplace is just as important.

In addition to carrying out their responsibilities efficiently this alignment, commonly referred to as "culture fit," makes sure that workers feel motivated, and connected so that they become a part of their teams and the whole company [12].

Improved teamwork, lower turnover rates, and higher employee engagement are just a few of the benefits that can result from a strong culture fit. However, a poor culture fit can lead to high attrition, workplace conflicts, and dissatisfaction, all of which can be expensive for businesses. These factors have made assessing cultural fit an essential part of the hiring process.

However, conventional techniques for determining cultural fit, such as personality tests, interviews, and intuition, are frequently subjective, tedious, and unreliable [4]. These techniques may unknowingly introduce bias by giving preference to applicants who "seem like a good fit" based more on personal preferences than on objective and unbiased ones.

Because hiring managers may unintentionally choose applicants who are similar to themselves or the current team, this subjectivity not only lessens the efficacy of culture fit assessments but also has the potential to restrict diversity within the company.

This study investigates a modern technique of evaluating cultural fit by utilizing artificial intelligence (AI) in order to overcome these obstacles. AI has the potential to revolutionize hiring procedures by making them less biased, faster, and more dependable [10, 11]. In particular, this study focuses on two AI models:

1. BERT (Bidirectional Encoder Representations from Transformers): This model is used for numerical matching of resumes and job descriptions to determine how well a candidate's experiences and qualifications match the job's requirements [3, 6].

2. Generative AI: This model provides in-depth qualitative analysis and richer insights by analyzing how applicants behave in response to behavioral questions and provides thor-

ough feedback regarding their personality, problem-solving abilities, and compatibility with the company's values and culture [9].

These tools provide a framework that can evaluate culture fit in a scalable and objective manner. This method gives recruiters a better grasp of each candidate's cultural fit by providing descriptive feedback in addition to numerical scores for ranking candidates. By addressing potential biases in the data and the AI models, the study also places a high priority on fairness and inclusivity.

Employers can enhance their hiring practices by using this hybrid approach which ensures that new hires are not only technically qualified but also culturally compatible, promoting long-term success for both the employee and the organization [12, 13]. The purpose of this paper is to thoroughly examine this framework, assess its efficiency, and demonstrate how it can overcome the limitations of culture fit evaluations.

II. Review of Literature

The alignment of a candidate's values, beliefs, and behaviors with the culture of the company has been a key component of effective hiring procedures. According to research, organizational performance, employee engagement, and satisfaction are all greatly impacted by culture fit [12]. However, because they favor consistency over diversity, traditional methods of evaluating cultural fit have frequently been criticized of promoting bias [13]. Organizations are increasingly adopting a "culture add" strategy in response to these concerns, emphasizing the hiring of applicants who share the company's core values and offer distinctive perspectives and abilities. This strategy encourages a dynamic and well-rounded workforce by increasing innovation and expanding organizational perspectives [13].

In the past, personality tests, interviews, and subjective assessments of a candidate's suitability were used to evaluate culture fit. It was difficult to obtain consistent and impartial results using these approaches since they frequently lacked standardization and were subject to biases [2]. Furthermore, the intricate and complex nature of organizational cultures was not taken into consideration by traditional methods, which resulted in mismatched hires that negatively affected team dynamics and productivity [18].

AI-driven techniques are being used in modern recruitment strategies to overcome these drawbacks because they allow for more scalable, objective, and data-driven evaluations of culture fit. These techniques have increased hiring efficiency while drastically lowering the subjectivity present in traditional techniques [2]. According to research, organizational performance, employee engagement, and satisfaction are all greatly impacted by culture fit [12].

Artificial Intelligence (AI) has completely changed how businesses assess culture fit in hiring. In order to detect semantic alignment in textual data, such as resumes and job descriptions, models such as BERT (Bidirectional Encoder Representations from Transformers) have proven remarkably effective [6]. Compared to conventional models, BERT is better at capturing contextual meanings because of its bidirectional text processing capabilities. BERT generates alignment scores, which are quantitative indicators of culture fit, by utilizing cosine similarity metrics [3, 6]. Although BERT is effective, one of its drawbacks is that it cannot identify subtle cultural indicators, which are often crucial in assessing a candidate's true match with an organization's culture [6].

These limitations have been overcome by generative AI models like GPT-3, which provide advanced capabilities for generating context-specific insights. Because of its few-shot learning capability, GPT-3 is able to produce comprehensive evaluations of candidates' answers to behavioral questions, offering complex assessments of their cultural fit [5, 9]. To add depth to the evaluation process, GPT-3 can examine how candidates' language reflects important qualities like adaptability or teamwork [9].

Significant ethical concerns are brought up by the use of AI in hiring, especially in relation to bias and fairness. Although they can lessen human subjectivity, predictive hiring algorithms can reproduce or magnify biases found in historical data [4]. Research highlights the significance of accountability, transparency, and frequent audits in guaranteeing the moral application of AI in hiring [8]. Frameworks such as AI4People offer practical suggestions to direct moral AI behavior, promoting ideas like beneficence, non-maleficence, and explicability to advance the welfare of society [7].

Additionally, there are differences in the global landscape of AI ethics, with the majority of guidelines coming from developed countries. This difference emphasizes the necessity of a range of viewpoints to guarantee that AI technologies are fair and inclusive in various cultural and economic contexts [8]. In order to create universally accepted ethical standards and address these differences, stakeholders including businesses, policymakers, and technologists must work together [8].

Companies such as Google and Unilever have effectively incorporated AI into their hiring procedures, leading to notable enhancements in productivity and diversity results [10]. AI systems are being used, for example, to evaluate candidates' behavioral responses, evaluate transferable skills, and automate resume screening. Communication, flexibility, and leadership are examples of transferable skills that are becoming more and more acknowledged as important markers of a candidate's capacity to succeed in a variety of roles [15].

AI-driven hiring tools are made even more effective by the idea of collaborative intelligence, which highlights the cooperation between humans and AI. Organizations can make more thoughtful and well-rounded hiring decisions by combining AI's analytical powers with the creative and intuitive strengths of humans [11]. Employees gain "fusion skills" from this collaboration, which helps them use AI tools efficiently in their workflows [11].

Effective hiring and employee engagement are based on a clearly defined corporate culture. Building a positive work environment requires components like fairness, respect, trust, and creativity [17]. These factors help to increase employee retention and productivity in addition to luring top talent [17].

Businesses are better positioned to maintain long-term success when they routinely evaluate and improve their culture through employee surveys and feedback [17].

Important information about how candidates fit with the organization's values can be gathered from behavioral interview questions. These insights provide a comprehensive picture of a candidate's compatibility when paired with AI-driven analysis, bridging the gap between conventional approaches and contemporary technologies [16]. To improve the accuracy of cultural assessments, AI tools such as Gemini, for instance, evaluate responses to evaluate qualities like cooperation and conflict resolution [16].

AI-powered solutions are changing the hiring process by tackling the biases and inefficiencies of conventional techniques. Recruiters can now quickly analyze large volumes of data thanks to BERT and GPT-3, which provide insights into both explicit qualifications and implicit cultural indicators [6, 9]. For these tools to be implemented effectively, organizations must balance ethical considerations and technological innovation. To ensure that AI advances organizational objectives while fostering equity and inclusivity, transparent procedures, frequent audits, and stakeholder cooperation are essential [4], [8].

Even though AI can revolutionize culture fit assessment, there are still obstacles to overcome, such as ethical concerns, the possibility of bias reinforcement, and the requirement that AI models be interpretable. To overcome these obstacles, AI technologies must be continuously improved, and diverse datasets must be used to guarantee fair results [4, 7, 8]. The development of AI models that can capture the complex nature of organizational culture should be the main goal of future research. This involves combining contextual and behavioral analysis to offer more in-depth understanding of candidates' compatibility. Furthermore, developing strong, equitable, and efficient hiring procedures will require promoting cooperation between AI systems and human assessors [11].

III. Objectives of the Study

The main goal of this study is to assess how well two advanced AI models BERT (Bidirectional Encoder Representations from Transformers) and Gemini (Generative AI) perform in determining cultural fit during the hiring process. Both models analyze three key inputs: resumes, job descriptions, and behavioral responses. Based on input data, including resumes, job descriptions, and behavioral responses, each model is used independently to compute results. The objective is to analyzing each model's effectiveness in determining candidates' and organizations' cultural alignment. to create and assess an AI-based system that assist companies in determining cultural fit during the hiring process.

The objectives of this study are outlined as follows:

1. **Make Culture Fit Evaluations More Objective:** Because traditional methods of evaluating culture fit frequently rely on subjective judgments or involuntary preferences, they can be biased and subjective. The goal of this research is to use AI to develop a more consistent and objective method of assessing how well candidates fit the culture, values, and working style of an organization.

2. **Evaluate using the BERT Model**

BERT model analyzes all three data sources as follows:

a. Resumes: BERT extracts relevant experiences, skills, and qualifications from resumes by analyzing their text.

b. Job Descriptions: It assesses job descriptions to determine the technical and cultural prerequisites for the position.

c. Responses to behavioral interview questions are processed by BERT, which finds semantic patterns consistent with the organization's values.

The objective is to determine how well candidates match the technical and cultural requirements of the position by calculating culture fit scores based on semantic similarity.

3. **Evaluate using the Gemini Model**

The Gemini model concentrates on producing context-sensitive insights while also analyzing behavioral responses, job descriptions, and resumes:

a. Resumes: Gemini assesses how well a candidate's background and abilities match the job description's organizational and cultural requirements.

b. Job Descriptions: It highlights the cultural elements that are incorporated into job descriptions, like the need for leadership, creativity, and teamwork.

c. Behavioral Reactions: Gemini provides thorough narrative feedback on candidates' communication, problem-solving, and adaptability skills.

The objective is to provide qualitative feedback regarding a candidate's fit with the company culture.

4. **Determine Strengths and Limitations of Each Model**

The study aims to investigate each model's strengths and limitations in relation to behavioral responses, job descriptions, and resumes:

BERT is expected to perform exceptionally well in processing vast volumes of textual data and in semantic analysis.

Gemini's capacity to produce contextual narratives is expected to yield deeper behavioral insights.

The study is able to recommend which model to use in particular recruitment scenarios by identifying these strengths and limitations.

5. **Provide Recruiters Separate, Actionable Results**

Every model produces its own outputs:

BERT: Offers a rapid quantitative measure of alignment by providing a numerical culture fit score.

Gemini: Generates in-depth narrative feedback that aids hiring managers in comprehending the finer details of a candidate's cultural fit.

In order to enable recruiters to use both tools independently for well-informed decision-making, the study attempts to provide them with these independent results. Using both quantitative and qualitative data, the objective is to assist recruiters in making well-informed decisions.

6. Maximize Recruitment Efficiency: The hiring process can be resource intensive and time consuming, particularly when it involves going through a lot of resumes and holding

1. Core Values (Top Priority)	2. Behavior/Attitude	3. Mission
4. Vision	5. Practices	6. Goals
7. Working Style	8. Diversity	9. Inclusion
10. Strategy	11. Work Life Balance	12. Policy (Least Priority)

Fig. 1. Culture Fit Dimensions Derived from Our Research

several rounds of interviews. By employing artificial intelligence (AI) to automate culture fit evaluations, the study seeks to speed up and improve this procedure.

7. Minimize Bias and Encourage Equity: The existence of unconscious bias, which can influence decision-making, is one of the main hiring challenges. This study attempts to reduce bias in the hiring process by utilizing AI models. In order to promote equity and inclusivity in the hiring process, AI can evaluate culture fit using data and patterns rather than subjective assessments.

8. AI's Performance in Culture Fit Evaluations: This study will also compare the AI system's evaluations with those of human recruiters to determine how well it performs in real-world scenarios. Finding out if the AI-driven strategy yields consistent, dependable, and comparable results to conventional methods is the goal. The accuracy and applicability of the AI-generated culture fit scores and narratives will be assessed using statistical techniques.

By accomplishing these goals, the study hopes to show that artificial intelligence (AI) can be a useful tool in the hiring process, offering more precise, equitable, and effective culture fit evaluations that eventually result in better hiring choices and more cohesive, strong teams.

IV. Proposed Framework

The study's conceptual framework shows how the BERT and Gemini models compute culture fit by processing the same three inputs resumes, job descriptions, and behavioral responses independently.

The distinct outputs produced by each model offer complementary insights into how well candidates match the job requirements and organizational culture. These models work independently to guarantee independent assessments, providing recruiters with a diversity of approaches to help them make informed choices.

From our extensive research, we have grouped realistic characteristics of culture into several dimensions/categories that are outlined in Figure 1.

The first step in the process is collecting input data, such as behavioral responses, resumes, and job descriptions.

1. Data Input Three primary inputs are used in the study, each of which represents a distinct aspect of a candidate's suitability:

- Resumes: Provide information about a candidate's work history, education, and abilities.
- Job descriptions: Describe the fundamental values, cultural norms, and technical requirements of the position.
- Behavioral Responses: Answers to behavioral interview questions that reveal a candidate's personality, problem-solving and teamwork skills.

After that, the data is pre-processed, which involves cleaning, dividing it into smaller components (tokens), and encoding it in a way that the AI models can understand.

2. Data Preprocessing The data is preprocessed to get it ready for analysis.

- Cleaning: Extra spaces and special characters that are not required are removed.
- Tokenization is the process of dividing the data into smaller units, or tokens, like words or phrases.
- Encoding: The text is transformed into embeddings, or numerical vectors, that the AI models can comprehend and process.

The study uses a priority scoring system to increase the accuracy of the assessment. When assessing culture fit, some factors such as behavior and core values are given more weight. For instance, a candidate who shares the company's basic beliefs is given preference over one who does the same. When evaluating candidates, this system makes sure that the most important cultural factors are given top priority.

Two categories of results are produced at the end of this process:

Qualitative outputs (narrative feedback) and quantitative outputs (culture fit scores). The culture fit score is a number that indicates how well a candidate fits the culture of the business and the position. The narrative feedback offers in-depth explanations of the candidate's conduct and compatibility with the company's basic principles. In order to help recruiters make more informed decisions

How does the Framework Operate: Input data includes behavioral responses, job descriptions, and resumes.

This data is processed independently by both models:

- BERT generates scores based on textual alignment and focuses on semantic similarity.
- Gemini produces behavioral insights by concentrating on contextual understanding.
- The results provides numerical scores and qualitative narratives to recruiters.
- These independent findings are used by recruiters to evaluate cultural fit and technical alignment, which facilitates in their hiring decisions.

V. Research Methodology

A. Data Collection

This study relies on a dataset obtained from Kaggle [1], a popular platform for data science and machine learning datasets. The chosen dataset includes job descriptions and resumes, which provide the textual data required for our culture fit assessment. The dataset is particularly valuable as it offers diverse examples of job postings and candidate profiles across different industries, roles, and experience levels, allowing for comprehensive testing and validation of our model.

The Kaggle dataset was selected due to its high quality and relevance for natural language processing tasks. Each entry in the dataset contains detailed job descriptions, which typically outline required skills, organizational values, and key responsibilities for specific roles. The dataset also includes sample resumes that detail candidates' educational backgrounds, work experiences, and professional skills. By structuring our data in this way, we can train and validate our culture fit model to assess alignment across multiple dimensions.

To prepare the data for analysis, we pre-processed the text by removing stop words, special characters, and performing tokenization to ensure that the model can effectively interpret the semantic meaning. Additionally, the dataset was split into training and test subsets to enable the evaluation of our model's performance on unseen data. This process ensures that the model's culture fit predictions are reliable and not merely reflective of patterns found in the training data. The dataset provides a solid foundation for our research, supporting the objective evaluation of job-candidate alignment through advanced NLP techniques.

B. Exploratory Data Analysis (EDA)

An essential phase in the data analysis process is exploratory data analysis (EDA), in which we look at and comprehend the data to find trends, patterns, and connections. EDA gives us a better understanding of the data's structure and properties and assists us in spotting any possible problems. This procedure is essential for directing additional research and enhancing the models we utilise for tasks such as predicting culture fit.

DA was applied to three primary categories of data in this study: 1. Job descriptions 2. Resumes 3. Behavioral Responses

The EDA in this study aims to determine how well the data fits the assessment of culture fit and to get it ready for analysis by AI models such as Generative AI and BERT.

The following steps were included in the EDA process for this study:

Preprocessing and Data Cleaning

Cleaning the data is the first step before analyzing it. This includes:

- Eliminating irrelevant text: Any unnecessary words or symbols that don't add to the analysis such as special characters or irrelevant resume sections eliminated.
- Managing missing data: In order to preserve the quality of the dataset, we either filled in any gaps in job descriptions, resumes, or responses, if feasible or eliminated them.
- Tokenization: This technique simplifies text for AI models to comprehend and analyze by breaking it up into smaller units known as tokens (words or phrases.

Understanding How Culture Fit Scores Are Distributed

Examining the distribution of culture fit scores using BERT's cosine similarity was the next stage in the EDA:

- Cosine Similarity Scores: The BERT model generates a score based on how closely a resume matches a job description. A better fit is indicated by a higher score, whereas a poorer fit is indicated by a lower score. The range of scores, showed how differently applicants fit the job descriptions. If the culture fit of particular job types was consistently higher or lower compared to others.

Analysis of Behavioral Response

Examining candidates' answers to behavioral questions which are used to evaluate cultural attributes like cooperation, problem-solving, and flexibility was another crucial component of EDA in this study. Which cultural characteristics were most valued across a range of job roles? For example, is teamwork crucial for all roles or just some? Each behavioral trait's average similarity score to job descriptions, which shows which attributes were most important for various roles. Whether the answers demonstrated how well the applicants aligned with the company's basic principles, which include creativity, teamwork, and flexibility.

Recognizing Patterns and Trends:

The EDA also aimed to identify any patterns or trends that might help guide the AI model:

- Patterns in Resumes and Job Descriptions: We looked at the frequency with which particular values or skills were mentioned in both job descriptions and resumes. For instance, we might discover that applicants who had a lot of experience working in teams were more likely to receive higher scores on the culture fit test for particular positions.
- Behavioral Trends: We searched for trends in the ways that various job categories (such as engineering, sales, and marketing) reacted to cultural characteristics by examining the behavioral responses. This made it easier for us to distinguish between characteristics that were more role-specific and those that were universally significant.

Identifying Outliers

Data points that deviate significantly from the rest of the data are known as outliers, They may point to issues or special situations that require more research. For example, it might be an anomaly if a candidate with a very different work history received a high culture fit score on a job description. By examining outliers, we were able to determine whether the AI models needed to be modified or provide an alternative explanation for specific data patterns.

Perspectives Particular to Roles

We were able to comprehend the differences in cultural fit between job categories by examining the data from a role-specific point of view. While independent problem-solving abilities may be more important in some roles, teamwork may be more important in others.

Some of the main conclusions were as follows:

• The wide range of alignment in the culture fit score distribution suggests that accurate AI models are required to distinguish between candidates who are a good fit and those who are not.

• Outliers were not found, so no additional modifications in the data preprocessing and model training phases is required. Behavioral traits like teamwork and adaptability were consistently valued across the majority of roles, underscoring the significance of analyzing these traits through AI-generated insights.

In conclusion, EDA was essential to understanding the structure of the data and getting it ready for the AI models to analyze it. It assisted in narrowing the study's focus, spotting possible problems, and directing the creation of the AI model to guarantee precise and practical culture fit evaluations.

C. Model Development and Analysis

This study employed artificial intelligence (AI) and data analytics to examine candidates' cultural fit based on a variety of data sources, including resumes, job descriptions, and behavioral responses. Assessing how well a candidate's qualifications, behavior, and organizational culture align is the aim of data analytics in this study.

The use of data analytics in this study is explained in detail below:

1. Using BERT to Analyze Semantic Similarity:

Cosine Similarity Calculation: The BERT model was used to determine the cosine similarity between resumes, job descriptions and behavioral responses as the first step in the data analytics process. The degree of similarity between these texts is measured by their cosine similarity. A score near 1 indicates high similarity, meaning the candidate's resume closely resembles the job description. The score ranges from 0 to 1. Low similarity is indicated by a score near zero, meaning that the candidate's resume does not match the job description. The input data is transformed by the BERT model into mathematical representations, or embeddings, that capture the context and meaning of the text. After that, cosine similarity is computed using these embeddings.

This analysis assists in determining how well a candidate's experiences and qualifications align with the job requirements, which is a crucial component of evaluating culture fit. For example, if a job description mentions that the applicant should be "innovative and collaborative," and the resume exhibits these qualities and contains comparable keywords, the cosine similarity score will be high, suggesting a good fit between the two cultures.

Analysis Steps: BERT uses its transformer-based architecture to transform text data into numerical embeddings. These embeddings' proximity is measured by cosine similarity, which yields a numerical culture fit score (from 0 to 1). By examining how closely related the words and phrases are in context, cosine similarity calculates how similar two texts are.

Stronger correspondence between the job description and the applicant's response or resume is indicated by a higher score. The average similarity score for applicants in all roles is known as the "mean culture fit score."

2. Generative AI (Gemini) Model for Qualitative and Quantitative Analysis:

The contextual alignment of behavioral responses and resumes with job descriptions is assessed by the Gemini model. It generates culture fit scores and also in-depth descriptive feedback. These scores represent their behavioral qualities (such as flexibility, problem-solving, or teamwork). The purpose of behavioral questions is to determine how well a candidate fits the culture of the company. We have carried out extensive research to understand which behavioral questions have a higher impact[19]. An example of a question designed to assess a candidate's ability to work with others is

"Describe a time when you worked as part of a team to solve a problem." Based on these answers, the AI produced narrative feedback that offered a more thorough examination of how well candidates' actions fit with the values of the company, including flexibility, problem-solving, and teamwork.

The additional layer of insight provided by this qualitative feedback goes beyond what is included in job descriptions and resumes.

Analysis Steps:

- Culture Fit Scores: numerical ratings for cultural traits like problem-solving, flexibility, and teamwork.
- Narrative Feedback: Detailed explanations of how a candidate's actions fit the values of the company.

Key characteristics (e.g., leadership, collaboration) were used to categorize input data. Using natural language generation to generate narrative feedback, the Gemini model evaluated how well these attributes aligned with the organization's core values.

Both sets of scores were gathered, and data analytics were used to examine how well candidates fit the company's culture and job roles overall.

D. Data Visualization

The graph depicted in Fig 2 illustrates a comparative analysis of BERT accuracy (scaled to a percentage) and Gemini scores for assessing cultural fit across seven job descriptions. Each bar represents the respective scores for a specific job, enabling a direct visual comparison of the quantitative and qualitative evaluation metrics.

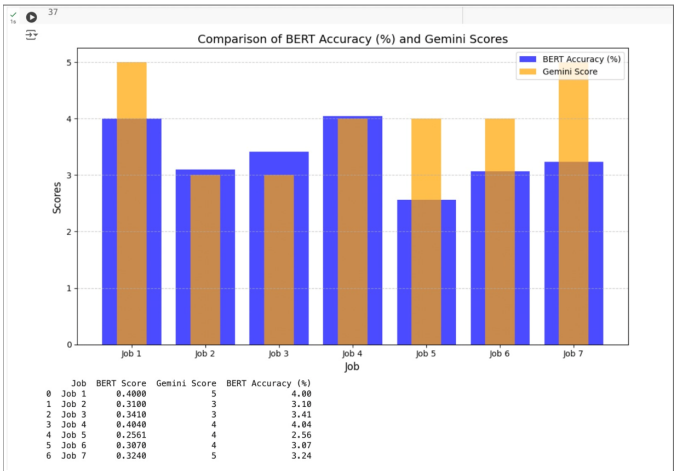

Fig. 2. Score Distribution Between Both Models

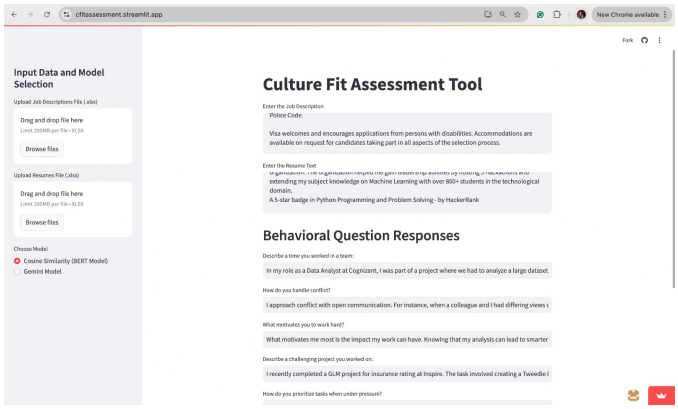

Fig. 3. BERT Model UI Representation

Comparison and Correlation:

- While BERT offers precise numerical matching, its range is relatively narrow, limiting its ability to capture deeper cultural nuances.
- Gemini provides richer contextual insights, as evidenced by its higher variability and alignment with qualitative cultural factors. Notably, Jobs 1 and 7 show both strong BERT accuracy and Gemini scores, highlighting alignment in both technical and cultural dimensions.

The visualization underscores the complementary nature of the BERT and Gemini models. While BERT quantifies technical fit effectively, Gemini enhances the assessment by addressing qualitative aspects. This dual approach ensures a holistic evaluation, offering significant potential for improving hiring decisions in real-world applications.

VI. Result and Comparison

The results of our study were derived from the deployment of two distinct Streamlit applications, each executing a different model for culture fit assessment: one based on the BERT model and the other utilizing a Generative AI model. This dual-application approach allowed us to directly compare the capabilities and limitations of each model and assess their effectiveness in providing accurate culture fit scores and insights.

BERT Model Application

The first application leveraged a BERT-based model for embedding and similarity computation. By transforming job descriptions, resumes, and behavioral question responses into high-dimensional vector representations, the BERT model calculated cosine similarity scores to estimate culture fit.

Although effective in identifying basic alignment between candidate documents and job requirements, the BERT model was somewhat limited in capturing complex, nuanced expressions of cultural values. It produced numerical scores that were informative but required interpretation to understand context-specific alignment fully.

Additionally, BERT's reliance on pre-trained embeddings sometimes led to generalizations, limiting its effectiveness in differentiating between subtle cultural aspects across different organizations.

Generative AI Model Application

The second application was designed to execute a Generative AI model, specifically optimized for generating detailed, context-sensitive culture fit assessments. This model was trained to not only compute alignment scores but also to provide narrative feedback that explains a candidate's fit with the company culture in a nuanced way. The Generative AI model offered a significant improvement over the BERT model in terms of comprehensiveness and interpretive accuracy. By generating descriptive feedback, the model provided insights into specific culture fit aspects, such as teamwork orientation, adaptability, and alignment with core company values, making it more user-friendly and effective in real-world recruitment scenarios.

The Generative AI model outperformed the BERT model in terms of accuracy and practical utility. By integrating contextual understanding, the Generative AI model could generate culture fit scores that closely aligned with human evaluators' assessments, as validated by comparative analyses with human-labeled data. Recruiters found the narrative insights generated by the Generative AI model particularly useful, as they helped highlight both strengths and potential areas for development in candidates, offering a holistic view of culture fit beyond a simple numerical score.

Overall Comparison and Insights

The results from both applications demonstrated the limitations of traditional similarity-based models, like BERT, when applied to tasks that require deeper contextual understanding and interpretation. The Generative AI model's ability to provide detailed explanations allowed for more nuanced evaluations, aligning closely with the needs of recruitment professionals. As such, the Generative AI model application proved to be a valuable tool for enhancing recruitment deci-

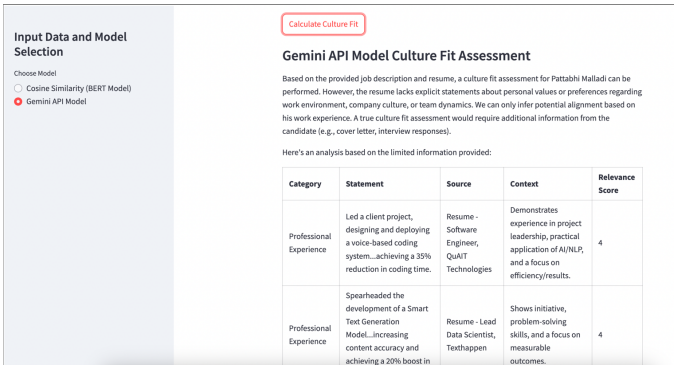

Fig. 4. Gemini Model Output

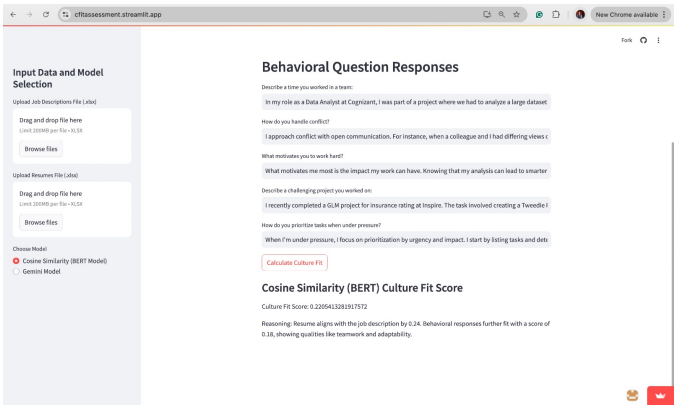

Fig. 5. BERT Model Output

sions by providing a more comprehensive and reliable culture fit assessment.

In conclusion, the dual-application approach underscored the importance of using advanced generative AI models in culture fit assessments, especially when understanding complex, multi-dimensional aspects of candidate compatibility with organizational values. While the BERT model application demonstrated foundational effectiveness, the Generative AI model emerged as a more powerful, insightful, and practical solution for culture fit analysis in recruitment contexts.

VII. Hypothesis Testing

Hypothesis testing is conducted in this study to understand whether the culture fit scores produced by the BERT and Gemini models differ significantly. When evaluating how well candidates fit with a company's culture, we want to know if these two AI models yield noticeably different results. Any assertion or presumption that can be verified with data is called a hypothesis. We have two categories of hypotheses in this case:

1. The null hypothesis (H_0)

According to the null hypothesis, the two models are not significantly different from one another. It's the notion that any difference between the BERT and Gemini scores is purely coincidental and not the result of one model's superiority in assessing cultural fit.

- In simpler words, The culture fit scores that the BERT and Gemini models provide do not differ significantly from one another. When it comes to evaluating culture fit, both models behave similarly or differently.

$$\mu_{\text{BERT}} = \mu_{\text{Gemini}}$$

2. Alternative Hypothesis (H_1)

According to the alternative hypothesis, the opposite is true. It suggests that there is a substantial difference between the two models, indicating that one of them—BERT or Gemini—offers a culture fit score that is superior to or more accurate than the other.

- In simpler words, The culture fit scores that the BERT and Gemini models provide differ significantly from one another. When evaluating culture fit, one model performs better than the other.

$$\mu_{\text{BERT}} \neq \mu_{\text{Gemini}}$$

- μ_{BERT}: Mean accuracy of the BERT model in evaluating culture fit.
- μ_{Gemini}: Mean accuracy of the Gemini model in evaluating culture fit.

Why Are These Hypotheses Tested?

Finding evidence to accept or reject the null hypothesis is the aim of testing these hypotheses. Rejecting the null hypothesis indicates that the differences between the models are substantial and genuine rather than merely coincidental. There is insufficient evidence to conclude that the models differ if the null hypothesis cannot be rejected.

Level of Significance (α)

A significance level, commonly represented by the symbol α, is used in hypothesis testing to determine when to reject the null hypothesis. Typically, the significance level is set at 0.05 (5%), meaning that we are prepared to accept a 5% chance of drawing the wrong conclusion.

- If the **p-value** from the test is **less than 0.05**, we reject the null hypothesis and accept the alternative hypothesis.
- If the **p-value** is **greater than 0.05**, we fail to reject the null hypothesis and conclude that there's no significant difference between the models.

Hypothesis Test Result

The culture fit scores produced by the two models (BERT and Gemini) are compared in this study using a two-sample t-test. This t-test determines whether there is a significant enough difference between the two groups' means (average scores) (BERT scores and Gemini scores) to rule out the null hypothesis.

Based on the probability value and t-statistic values calculated using python in Fig 6, We determine that there **is a statistically significant difference in the two models' scores** and that they do differ in their evaluations of culture fit as the p-value is less than 0.05.

```python
bert_scores = [
    4.31, 2.59, 2.59, 0.86, 6.90, 1.72, 2.59, 6.90, 6.90, 3.45, 3.45, 4.31, 5.17, 4.31, 6.04, 1.72,
    6.04, 8.62, 6.04, 0.86, 6.04, 4.31, 2.59, 2.59, 3.45, 1.72, 1.72, 0.86, 4.31, 3.45, 8.62, 1.72,
    2.59, 6.04, 6.04, 3.45, 4.31, 4.31, 5.17, 7.76, 1.72, 4.31, 2.59, 3.45, 8.62, 5.17, 6.90, 0.86,
    8.62, 6.90, 6.04, 0.86, 1.72, 2.59, 0.86, 7.76, 6.90, 1.72, 7.76, 6.04, 7.76, 5.17, 5.17, 3.45,
    8.62, 8.62, 1.72, 6.04, 4.31, 5.17, 1.72, 4.31, 8.62, 3.45, 7.76, 5.17, 1.72, 0.86, 4.31, 7.76,
    5.17, 8.62, 6.04, 4.31, 8.62, 5.17, 4.31, 7.76, 5.17, 6.90, 1.72, 7.76, 4.31, 6.04, 8.62, 0.86,
    8.62, 4.31, 8.62, 1.72, 5.17, 0.86, 2.59, 4.31, 3.45, 2.59, 8.62, 4.31, 6.90, 3.45, 0.86, 6.04,
    6.90, 2.59, 5.17, 1.72, 6.04, 6.90, 5.17, 5.17, 4.31, 6.90, 0.86, 8.62, 3.45, 3.45, 2.59, 2.59,
    3.45, 0.86, 1.72, 7.76, 8.62, 1.72, 4.31, 7.76, 7.76, 6.90, 0.86, 0.86, 7.76, 6.90, 8.62, 7.76,
    6.04, 2.59, 2.59, 1.72, 4.31, 2.59, 5.17, 1.72, 2.59, 4.31, 0.86, 0.86, 8.62, 7.76, 6.04, 0.86,
    6.90, 2.59, 6.90, 4.31, 0.86, 6.04, 4.31, 5.17, 5.17, 6.90, 7.76, 1.72, 7.76, 8.62, 7.76, 8.62,
    8.62, 6.04, 0.86, 6.04, 8.62, 4.31, 5.17, 8.62, 6.04, 4.31, 6.04, 7.76, 2.59, 6.90, 1.72, 6.04
]

gemini_scores = [
    5, 3, 3, 4, 6, 4, 5, 6, 6, 4, 5, 5, 6, 5, 7, 4, 6, 8, 6, 3, 6, 5, 3, 3, 4, 4, 4, 3, 5, 4, 8, 4,
    4, 6, 6, 4, 5, 5, 6, 8, 4, 5, 3, 4, 8, 6, 6, 3, 8, 6, 6, 3, 4, 4, 3, 8, 6, 4, 8, 6, 8, 6, 6, 4,
    8, 8, 4, 6, 5, 6, 4, 5, 8, 4, 8, 6, 4, 3, 5, 8, 6, 8, 6, 5, 8, 6, 5, 8, 6, 6, 4, 8, 5, 6, 8, 3,
    8, 5, 8, 4, 6, 3, 4, 5, 4, 3, 8, 5, 6, 4, 3, 6, 6, 4, 6, 4, 6, 6, 6, 6, 5, 5, 3, 8, 4, 4, 3, 3,
    4, 3, 4, 8, 8, 4, 5, 8, 8, 6, 3, 3, 8, 6, 8, 8, 6, 3, 3, 4, 5, 3, 6, 4, 3, 5, 3, 3, 8, 8, 6, 3,
    6, 4, 6, 5, 3, 6, 5, 6, 6, 6, 8, 4, 8, 8, 8, 8, 8, 6, 3, 6, 8, 5, 6, 8, 6, 5, 6, 8, 3, 6, 4, 6
]

from scipy.stats import ttest_ind

# Performing a two-sample t-test
t_stat, p_value = ttest_ind(bert_scores, gemini_scores)

print("T-statistic:", t_stat)
print("P-value:", p_value)

# Significance level
alpha = 0.05

if p_value < alpha:
    print("Reject the null hypothesis (H0). Gemini scores are significantly different.")
else:
    print("Fail to reject the null hypothesis (H0). No significant difference found.")
```

```
T-statistic: -2.791449976282911
P-value: 0.005510743220961518
Reject the null hypothesis (H0). Gemini scores are significantly different.
```

Fig. 6. Hypothesis Statistics and Result

VIII. Conclusion

This study concludes by showing the benefits of evaluating culture fit in hiring through a hybrid AI-driven approach. Assessing culture fit has previously been a subjective procedure that frequently depends on human judgment in personality tests or interviews. These approaches could be biased and inconsistent. However, this study provides a more objective, effective, and data-driven method of evaluating culture fit by using BERT and Gemini.

AI in hiring can result in better, more informed hiring decisions, according to the findings of the BERT and Gemini models. Both models generated culture fit score, but Gemini offered more in-depth analysis with descriptive feedback on how well candidates' behaviors and values matched the company's culture. Recruiters were able to compare candidates more easily with the use of the scores generated. Gemini's narrative feedback offered a deeper level of understanding than just numbers, enabling recruiters to comprehend the logic behind culture fit scores. Utilizing multiple AI models for a more thorough assessment is recommended by hypothesis testing, which revealed a statistically significant difference between the BERT and Gemini models' culture fit scores.

In conclusion, this hybrid approach improves the hiring process by decreasing bias, increasing efficiency, and offering a more thorough understanding of a candidate's possible fit with the organization. Employers can make more data-driven choices and hire applicants who not only have the necessary skills but also fit with the company's values and culture by implementing AI-based techniques like BERT and Gemini. This strategy can result in a more friendly workplace, increased retention rates, and improved employee satisfaction. It can also provide a foundation for future advancements in AI-driven recruitment, making hiring processes more fair, transparent, and effective.

Acknowledgment

We would like to express our deepest gratitude to Dr. Stephen Wheeler, Clinical Assistant Professor at the University of North Texas, Ravi Varma Kumar Bevara, and Krishna Annavaram, our Teaching Assistants, for their invaluable guidance and support throughout this research.

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